College of Engineering Pune
(An Autonomous Institute of the Govt. of Maharashtra)

SUBJECT: SURVEYING AND GEOMATICS
UNIT 5

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Unit 5

Photogrammetry

Objects, applications to various fields, aerial camera, comparison of map & vertical photograph, vertical tilted and oblique photographs, scale of vertical photograph, Mirror Stereoscope, photo interpretation, etc. Geographic Information System (GIS): definition and meaning, data modes for GIS, components of GIS, and application to Civil Engineering, etc.

PHOTOGRAMMETRY

Photogrammetry is the art, science, and technology of obtaining reliable information about physical objects and the environment through processes of recording, measuring, and interpreting photographic images and patterns of recorded radiant electromagnetic energy and other phenomena

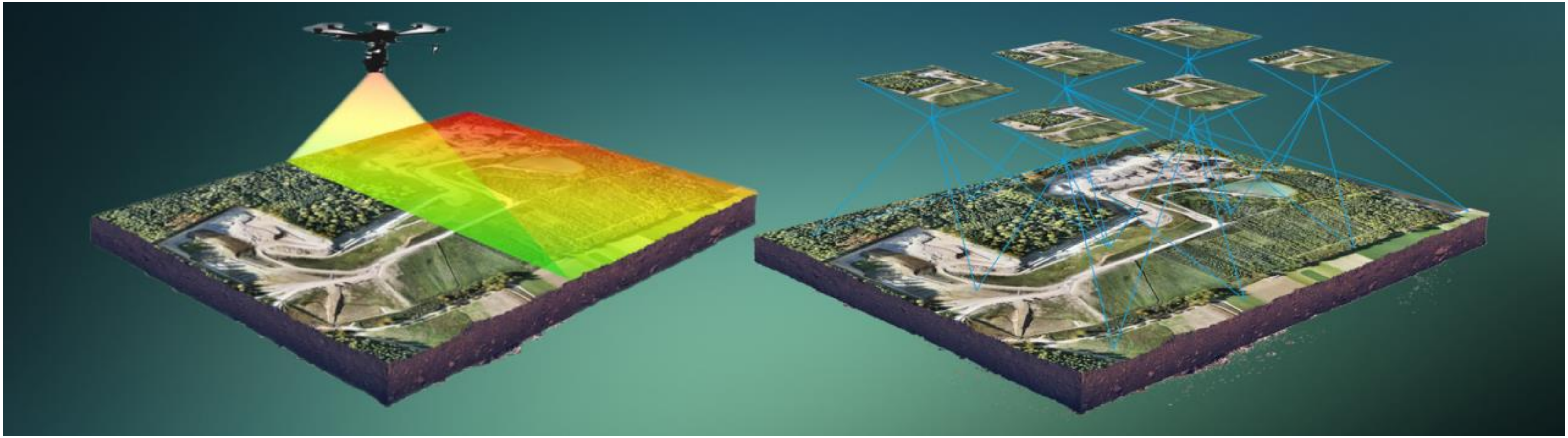
(Wolf and Dewitt, 2000; McGlone, 2004).

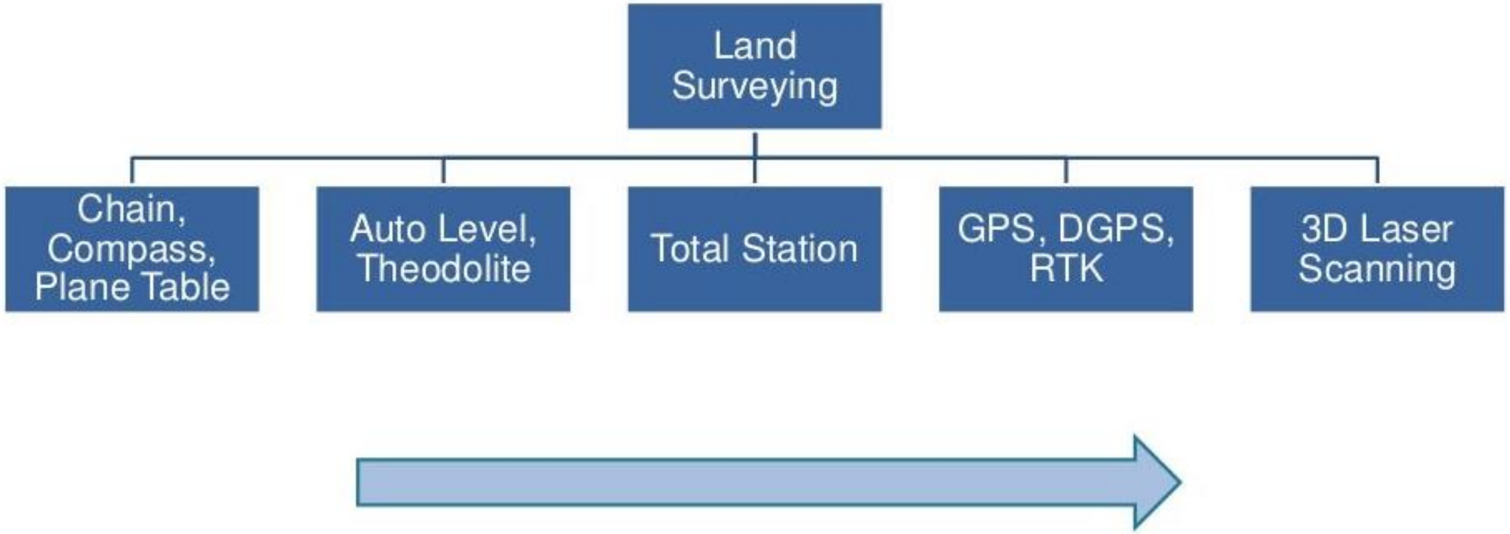
"The acquisition of physical data of an object without touch or contact"
- Lintz and Simonett, 1976

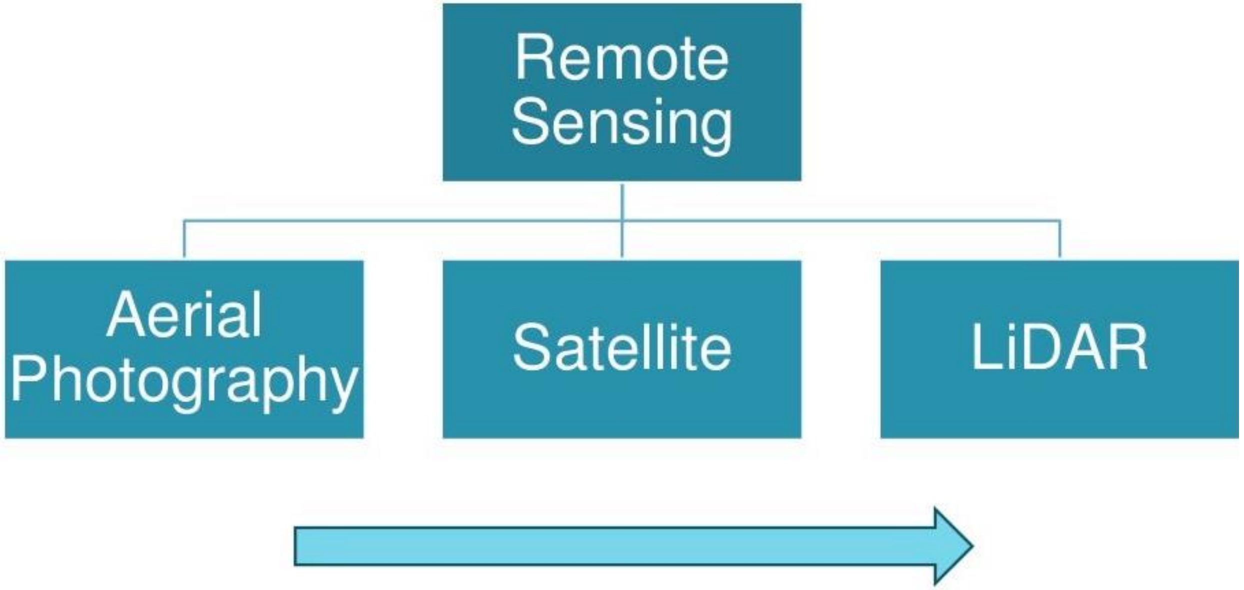
"The observation of a target by a device some distance away"
- Barrett and Curtis, 1982

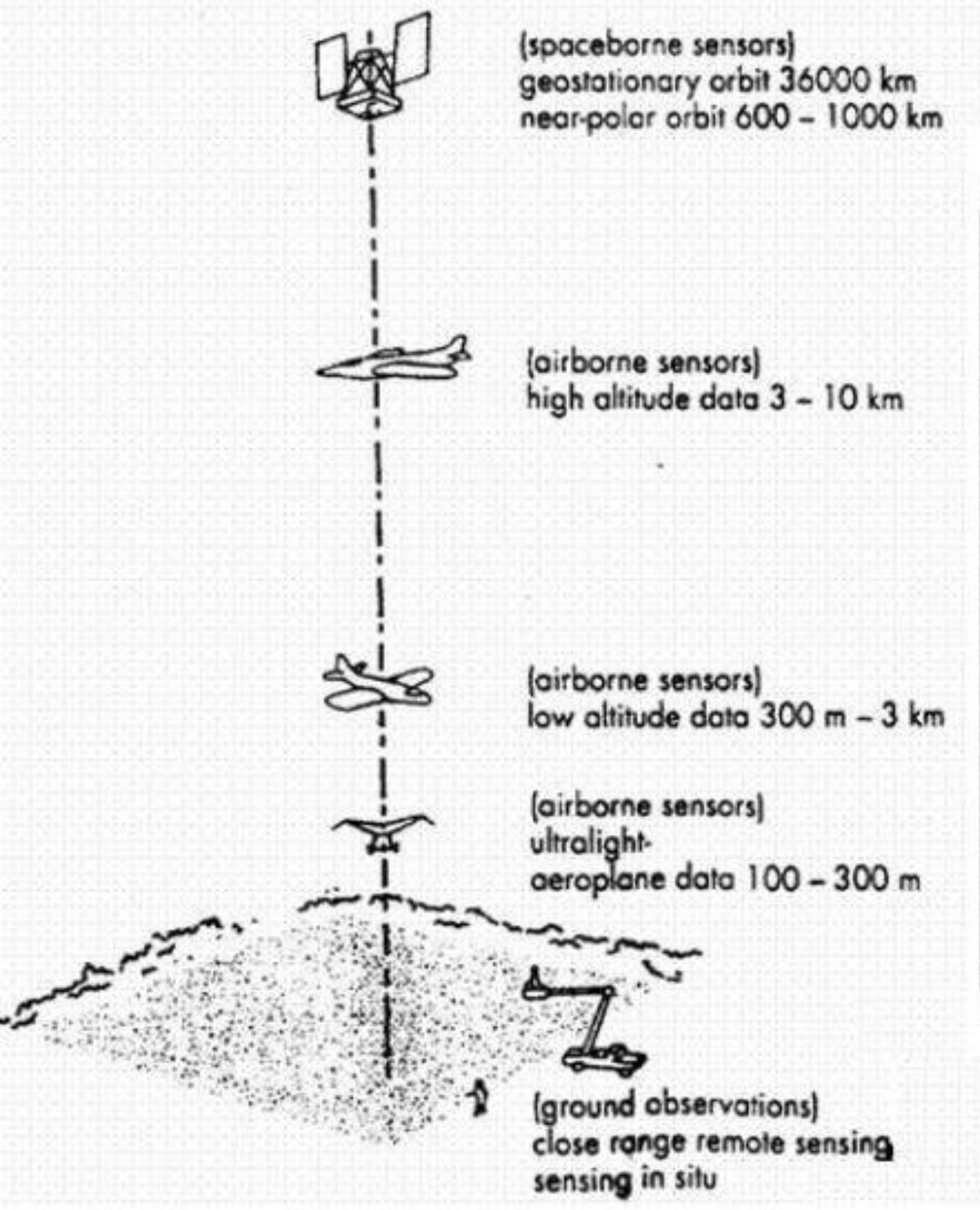
"The use of electromagnetic radiation sensors to record images of the environment, which can be interpreted to yield useful information" **- Curran, 1985**

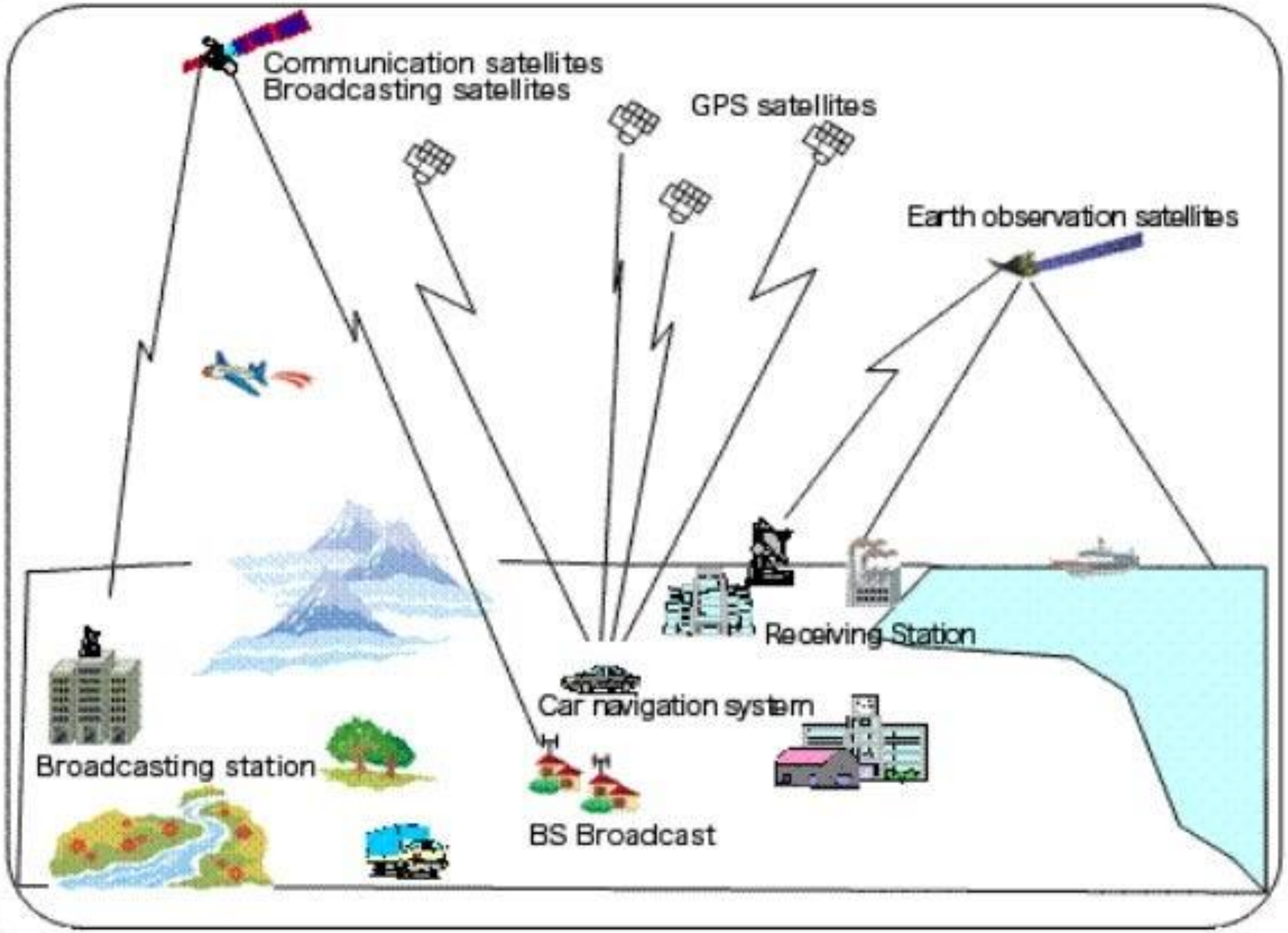
"The use of sensors, normally operating at wavelengths from the visible to the microwave, to collect information about the Earth's atmosphere, oceans, land and ice surfaces" **- Harris, 1987**




Land Surveying
Chain, Compass, Plane Table
Auto Level, Theodolite
Total Station
GPS, DGPS, RTK
3D Laser Scanning

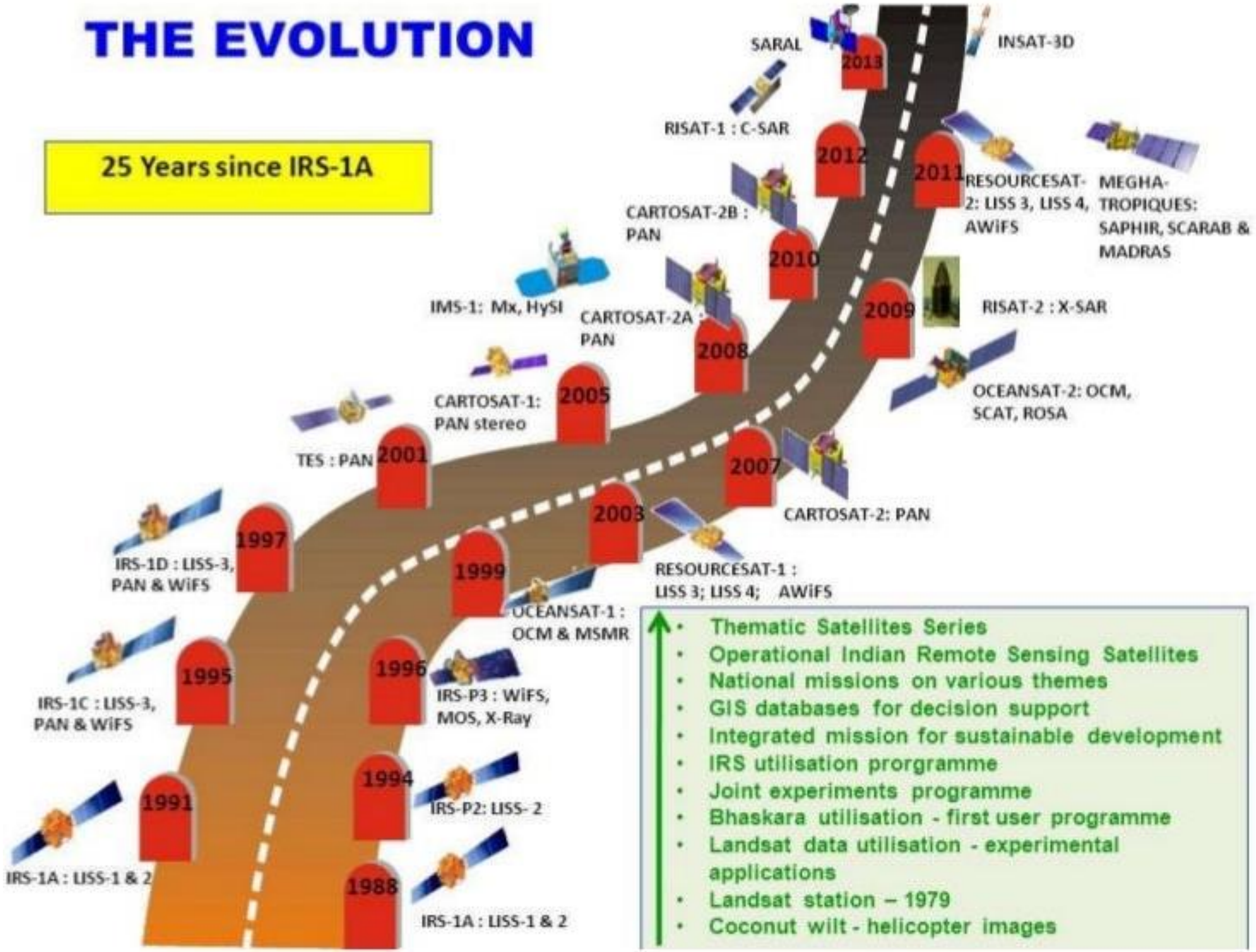

Remote Sensing
Aerial Photography
Satellite
LiDAR


(spaceborne sensors)
geostationary orbit 36000 km
near-polar orbit 600 – 1000 km
(airborne sensors)
high altitude data 3 – 10 km
(airborne sensors)
low altitude data 300 m – 3 km
(airborne sensors)
ultralight-
aeroplane data 100 – 300 m
(ground observations)
close range remote sensing
sensing in situ


Communication satellites
Broadcasting satellites
GPS satellites
Earth observation satellites
Receiving Station
Car navigation system
Broadcasting station
BS Broadcast

THE EVOLUTION

25 Years since IRS-1A



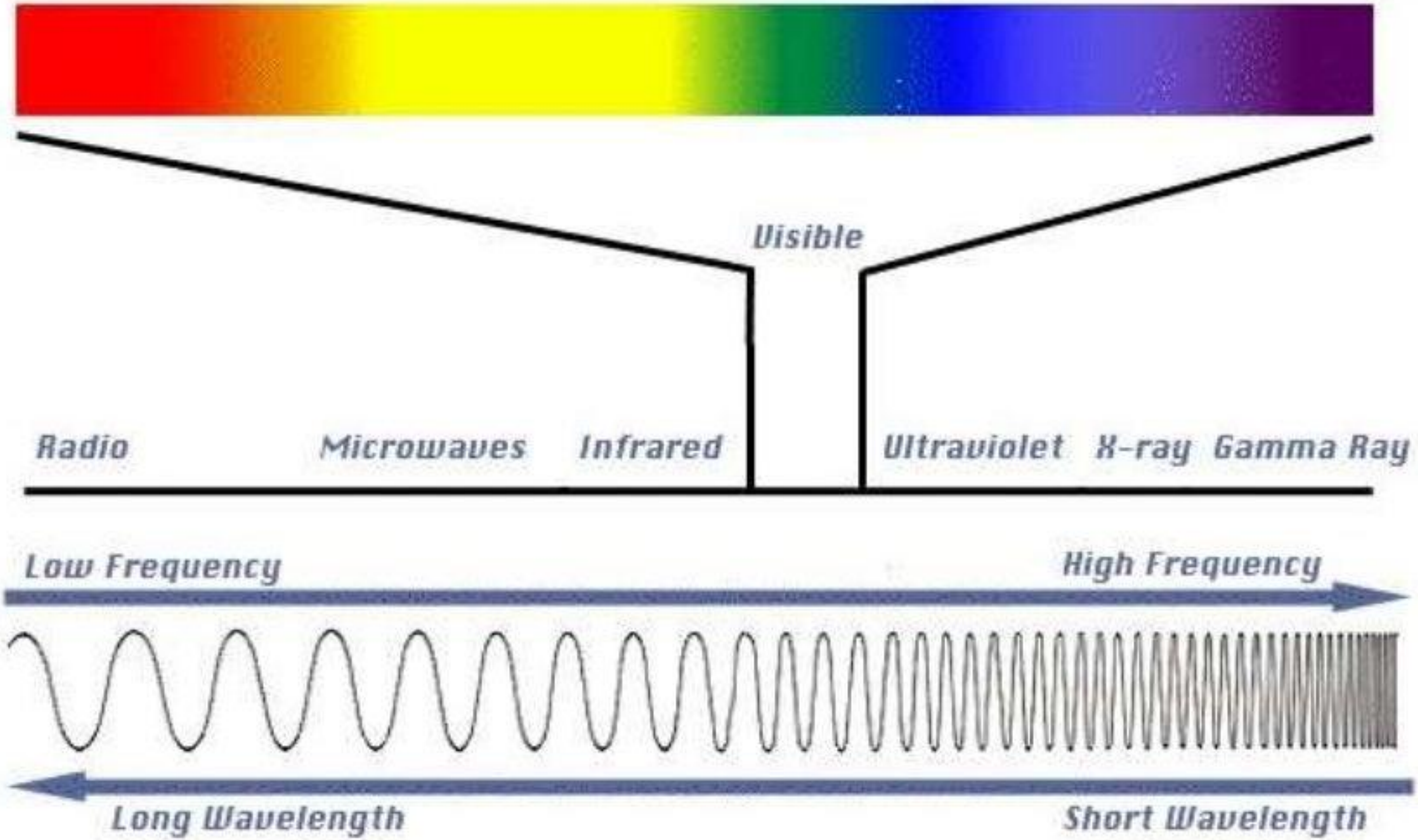
- Thematic Satellites Series
- Operational Indian Remote Sensing Satellites
- National missions on various themes
- GIS databases for decision support
- Integrated mission for sustainable development
- IRS utilisation prorgramme
- Joint experiments programme
- Bhaskara utilisation - first user programme
- Landsat data utilisation - experimental applications
- Landsat station – 1979
- Coconut wilt - helicopter images

1000 Applications & Uses

67 Major Applications

LISS – III Camera

Device to sample and measure radiation (sensor)



Medium = Electro Magnetic Radiation (EMR)

Introduction to Photogrammetry

Definition of Photogrammetry:

The art, science, and technology of obtaining information about physical objects and the environment by photographic and electromagnetic images.

Categories:

1. **Metrical Photogrammetry:** obtaining measurements from photos from which ground positions, elevations, distances, areas, and volumes can be computed and topographic or planimetric maps can be made.
2. **Photo Interpretation:** evaluation of existing features in a qualitative

PRINCIPLE OF PHOTOGRAMMETRY

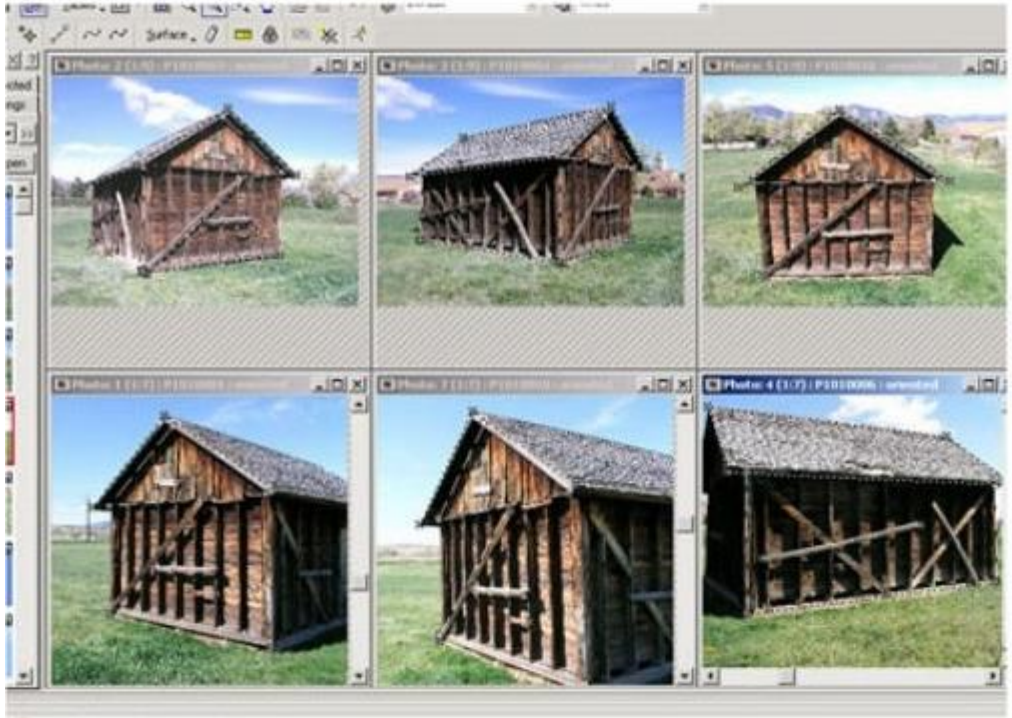
- Principle of photogrammetric survey in its simplest form is very similar to that of the plane table survey.
- Only difference is that the most of the work which in plane table survey is executed in the field, is done in office.
- The principal point of each photograph is used as a fixed station and rays are drawn to get points of intersections very similar to those used in plane table.
- Is suitable for topographical or engineering surveys and also for those projects demanding higher accuracy.
- It is unsuitable for dense forest and flat-sands due to the difficulty of identifying points upon the pair of photographs.

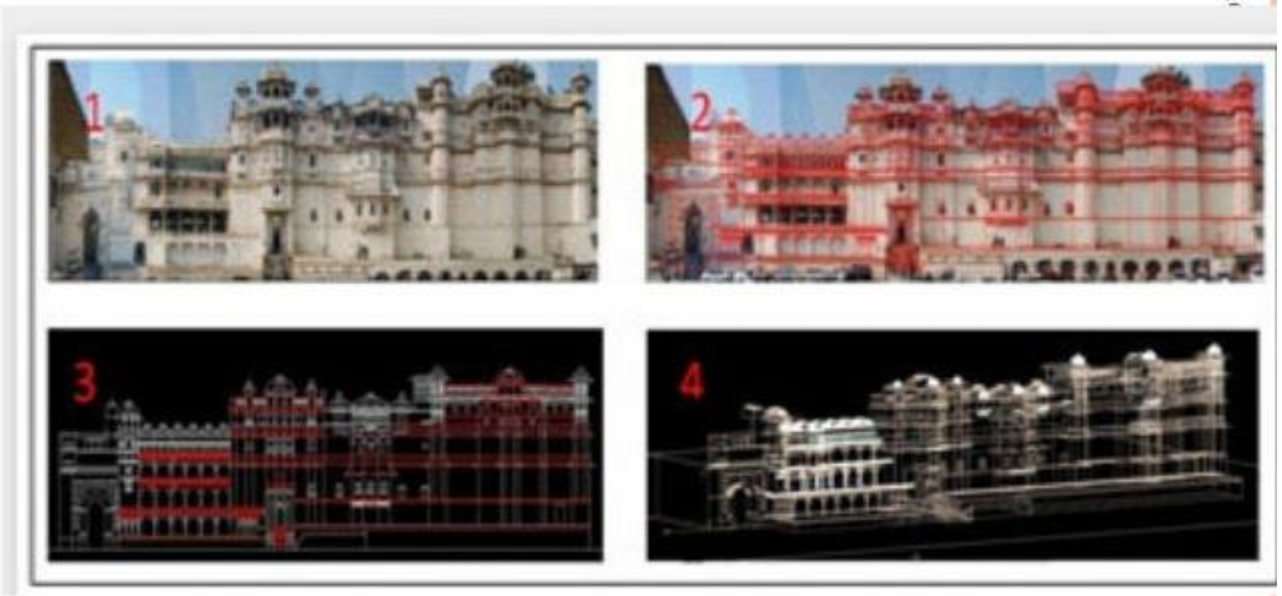
Types of Photogrammetry

1. **Aerial** – series of photographs of an area of terrain in sequence using a precision camera.
2. **Terrestrial** – photos taken from a fixed and usually known position on or near the ground with the camera axis horizontal or nearly so.
3. **Close range** – camera close to object being observed. Most often used when direct measurement is impractical.

TERRESTRIAL PHOTOGRAPHS

- photographs are taken from elevated ground stations.
- Method is very similar that the camera is in stationary position.
- Camera used in this method is called photo-theodolite as it will require t same features as theodolite.
- It is much cheaper and can be carried out by individual surveying firms also.





Er. Pramesh Ha

TERRESTRIAL PHOTOGRAPHS

- Difference between this and plane tabling is that more details are at once obtained from the photographs and their subsequent plotting etc. is done by the office while in plane tabling all the detailing is done in the field itself.
- Fig A and B are the two stations at the ends of base AB.
- Arrows indicate the directions of horizontal pointing (in plan) of the camera.
- For each pair of pictures taken from the two ends, the camera axis is kept parallel to each other.
- From economy and speed point of view, minimum number of photographs should be used to cover the whole area and to achieve this it is essential to select the best positions of the camera stations.
- Study of the area should be done from the existing maps, and a ground reconnaissance should be made. Selection of actual stations depends upon the size and ruggedness of the area.
- These photographs provides the front view of elevation & are generally used for the survey of structure & Architectural Monuments.

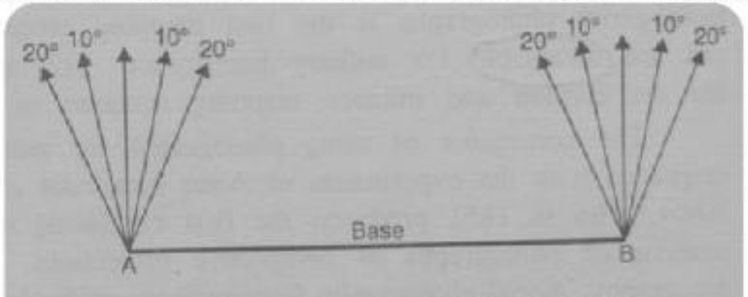


FIG. 14.1. DIRECTION OF POINTINGS IN TERRESTRIAL PHOTOGRAMMETRY

Photogrammetry for Engineering

Photogrammetry is the process of measuring images on a photograph.

Modern Photogrammetry also uses radar imaging, radiant electromagnetic energy detection and x-ray imaging – called **R*emote Sensing*.**

Origins of Remote Sensing

Remote Sensing began with Aerial Photography

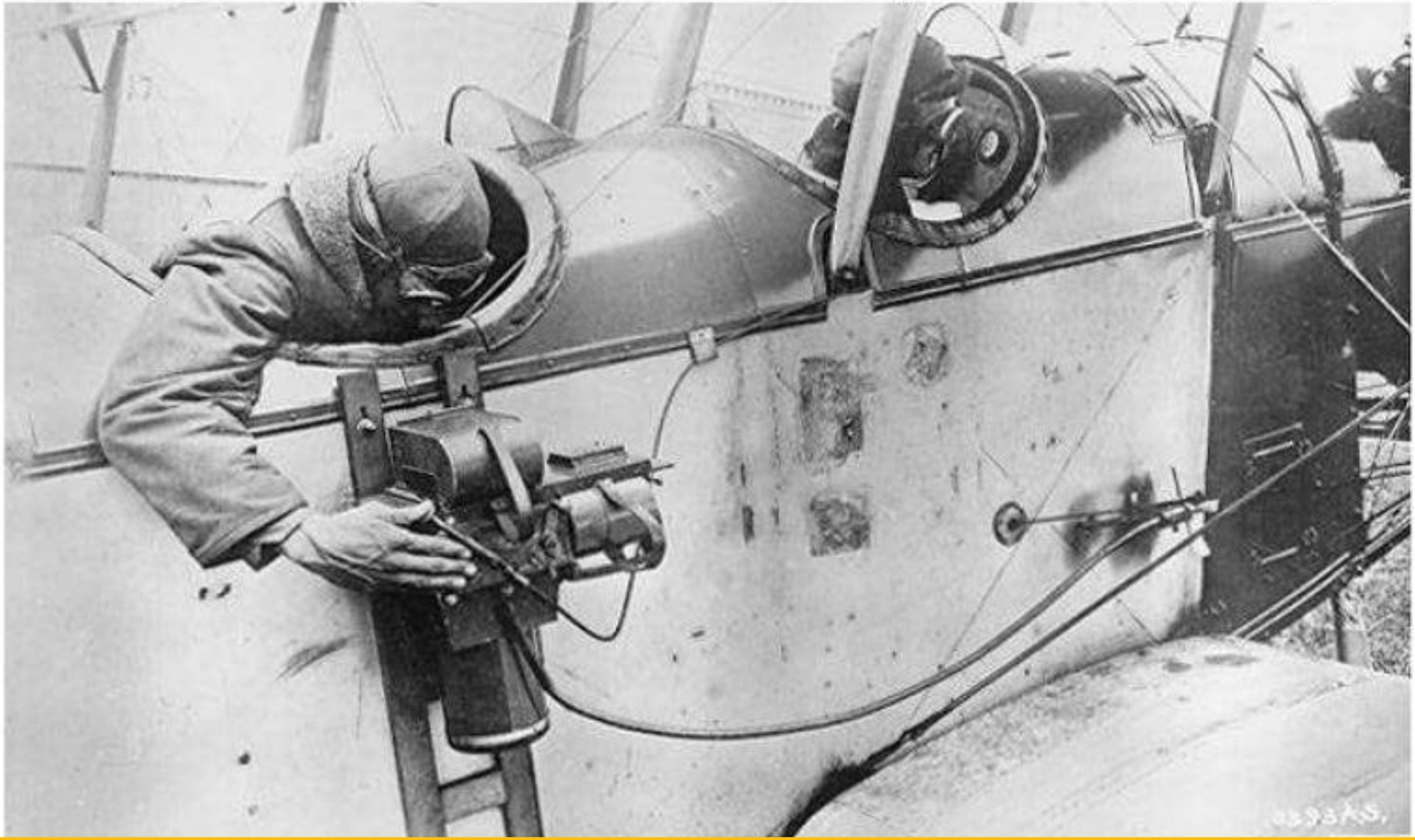
First photographs taken in 1839

In1858 Gasper Felix Tournachon "Nadar" takes photograph of village of Petit Bicetre in France from a balloon

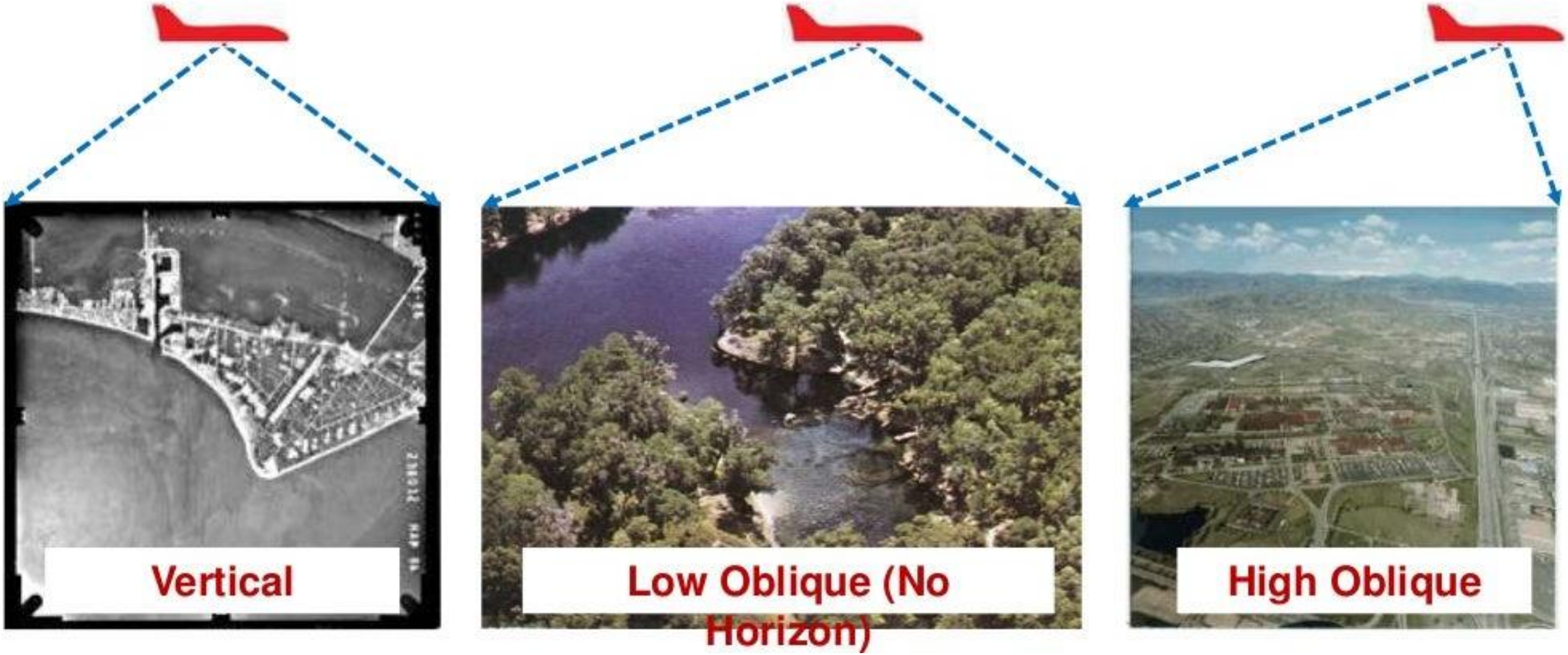
Paris by Nadar, circa 1859

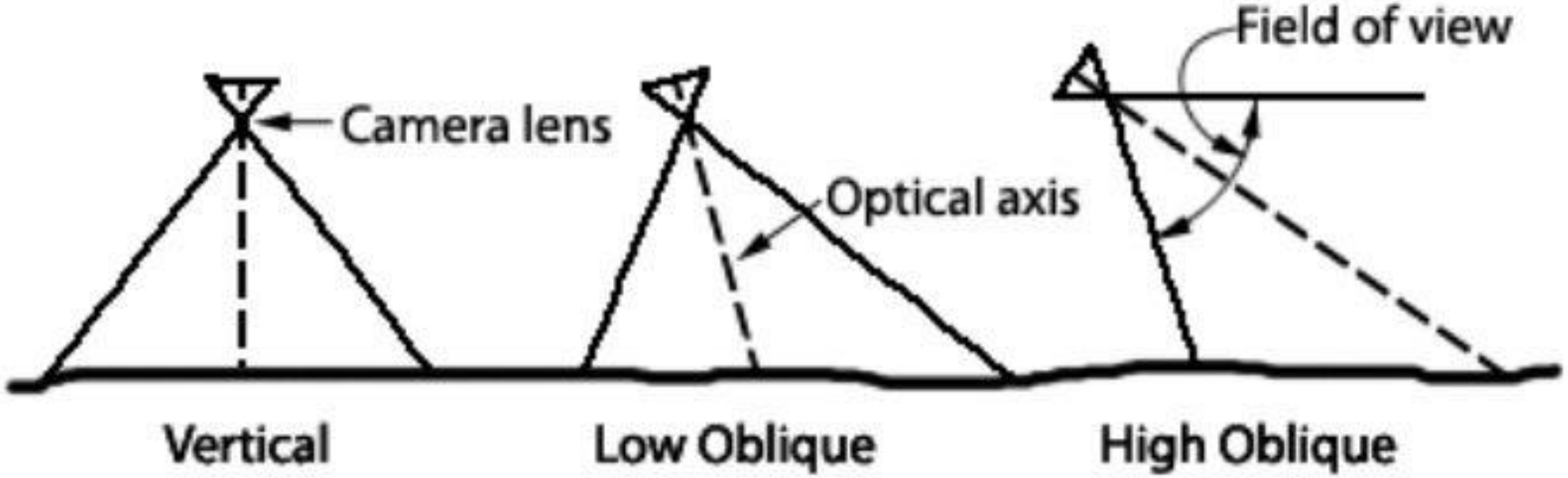
Boston by Black and King (1860

World War One was a major impetus to development of aerial photography

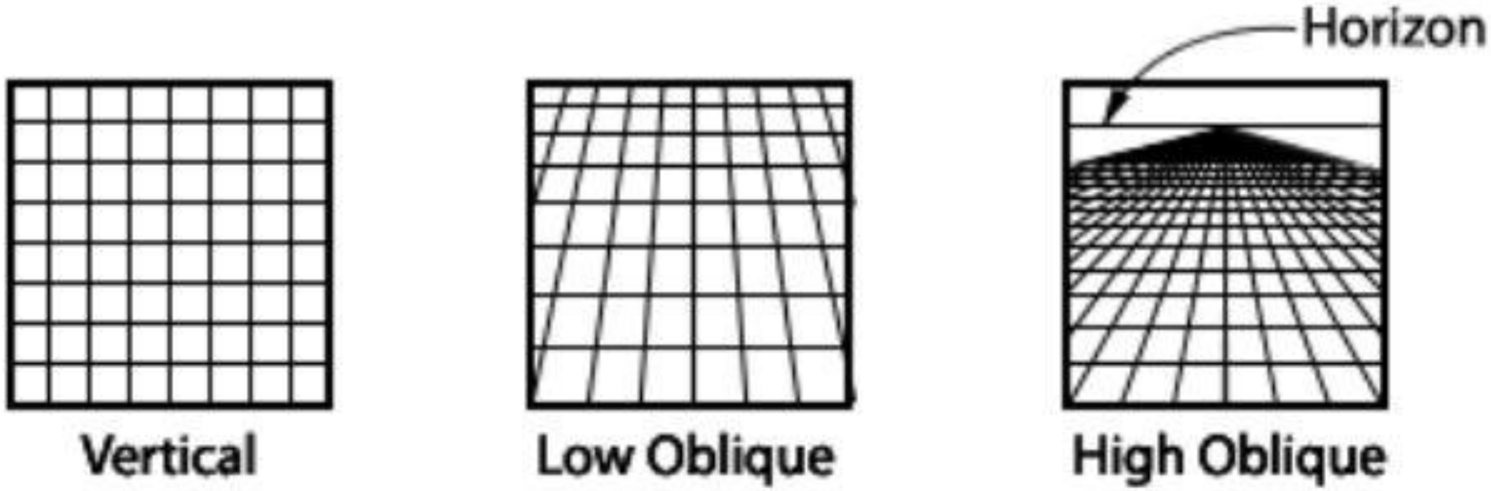


Types of Aerial Photographs





Camera orientation for various types of aerial photographs



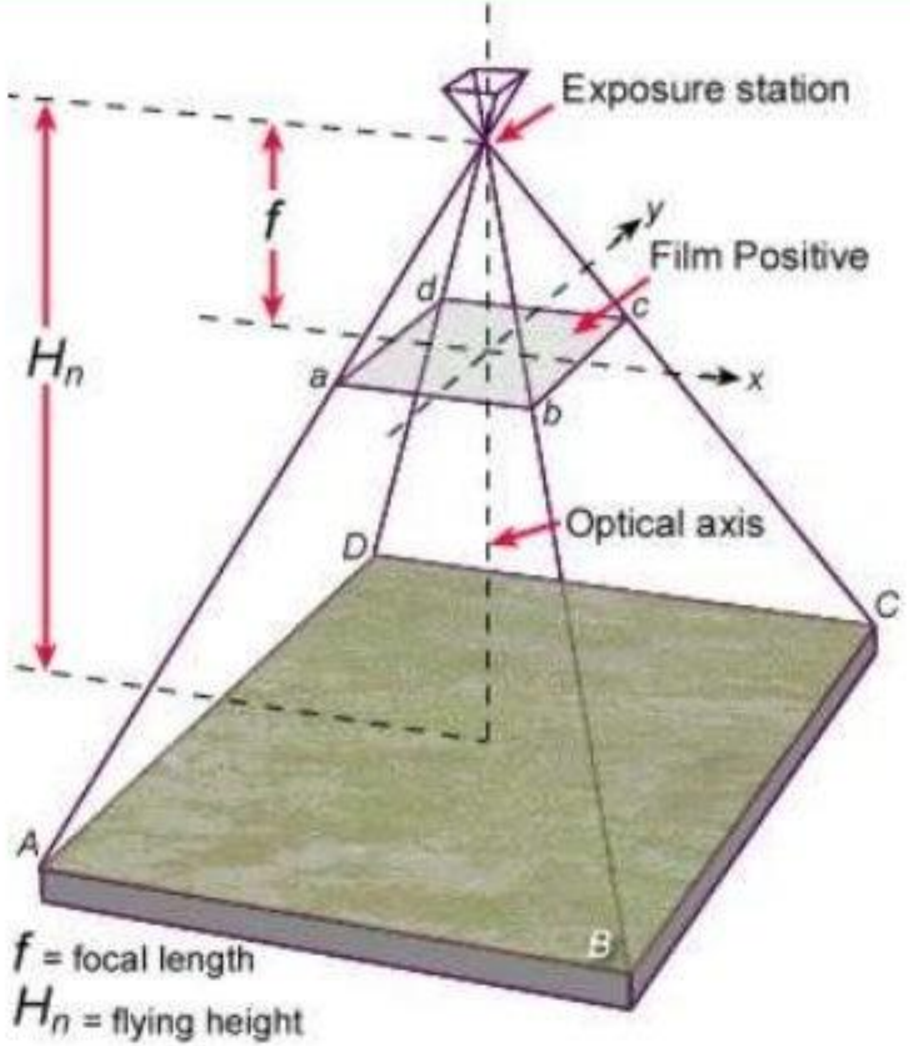
How a grid of section lines appears on various types of photos.

Vertical Aerial Photograph

Vertical is most important as it has minimum distortion and can be used for taking measurements.

Characteristics:

- Tilt <= 3^0 from the vertical
- Scale is approximately constant throughout the photo
- Most common format is 9” x 9” photograph

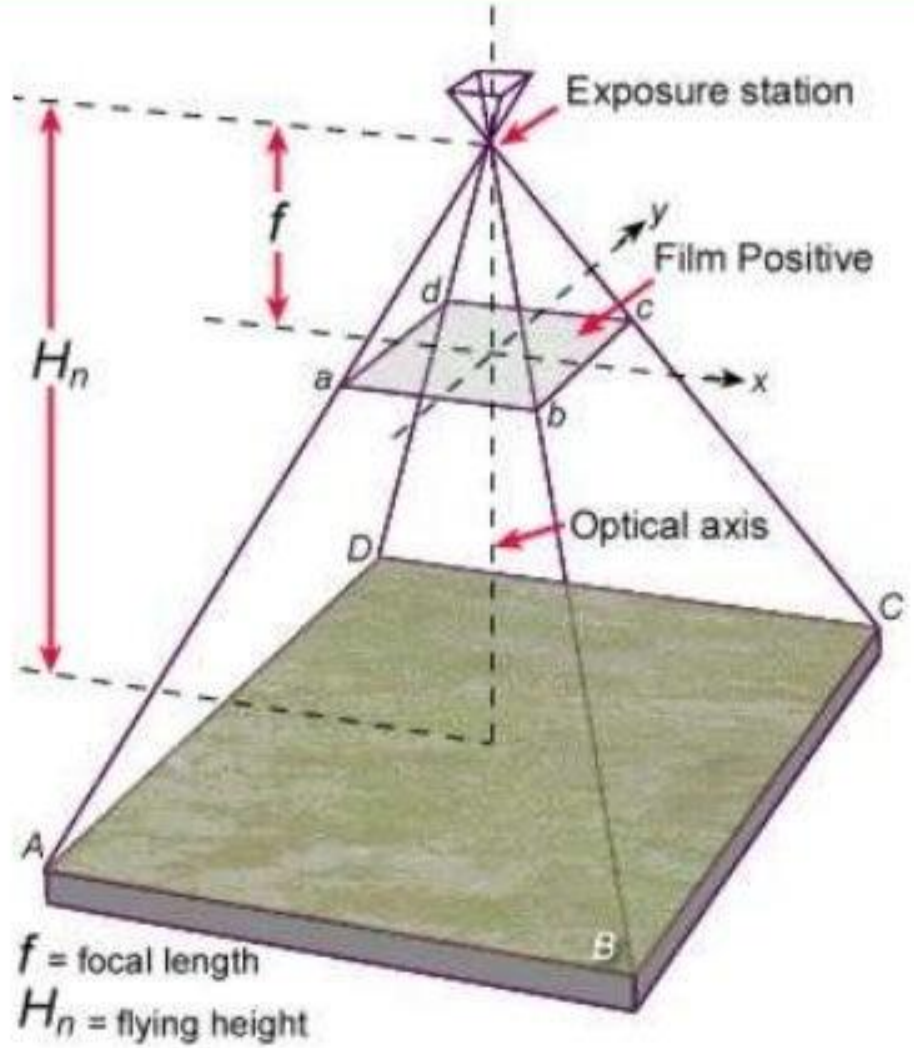


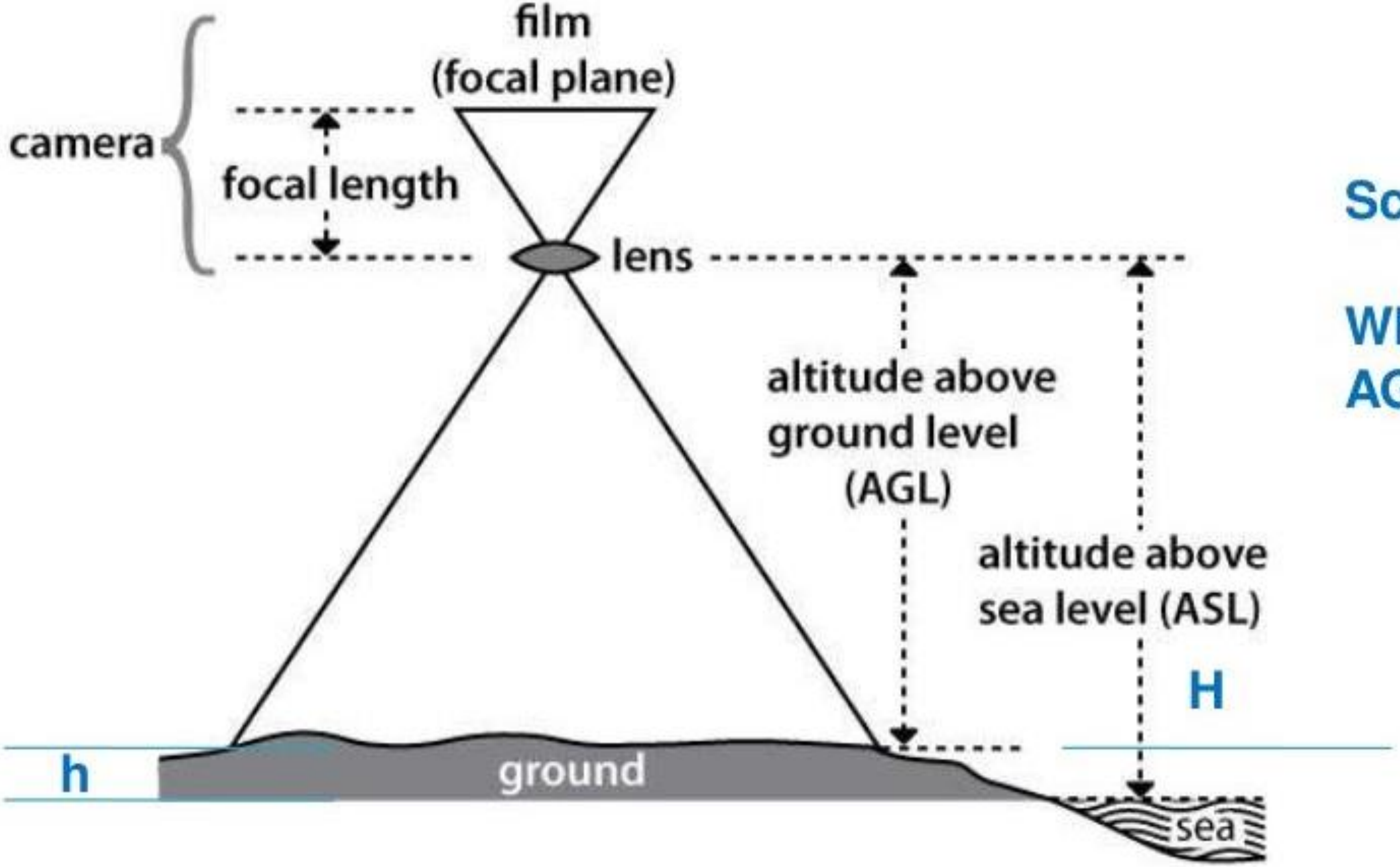
Scale Measurement

If you know focal length of camera and height of aircraft above the ground you can calculate the scale of the photograph.

Scale = f/H-h

f = focal length (distance from centre of lens to film surface)
H = flying height of aircraft above sea level
h = height of ground above sea level





Scale = focal Length/AGL

Where
AGL = H-h

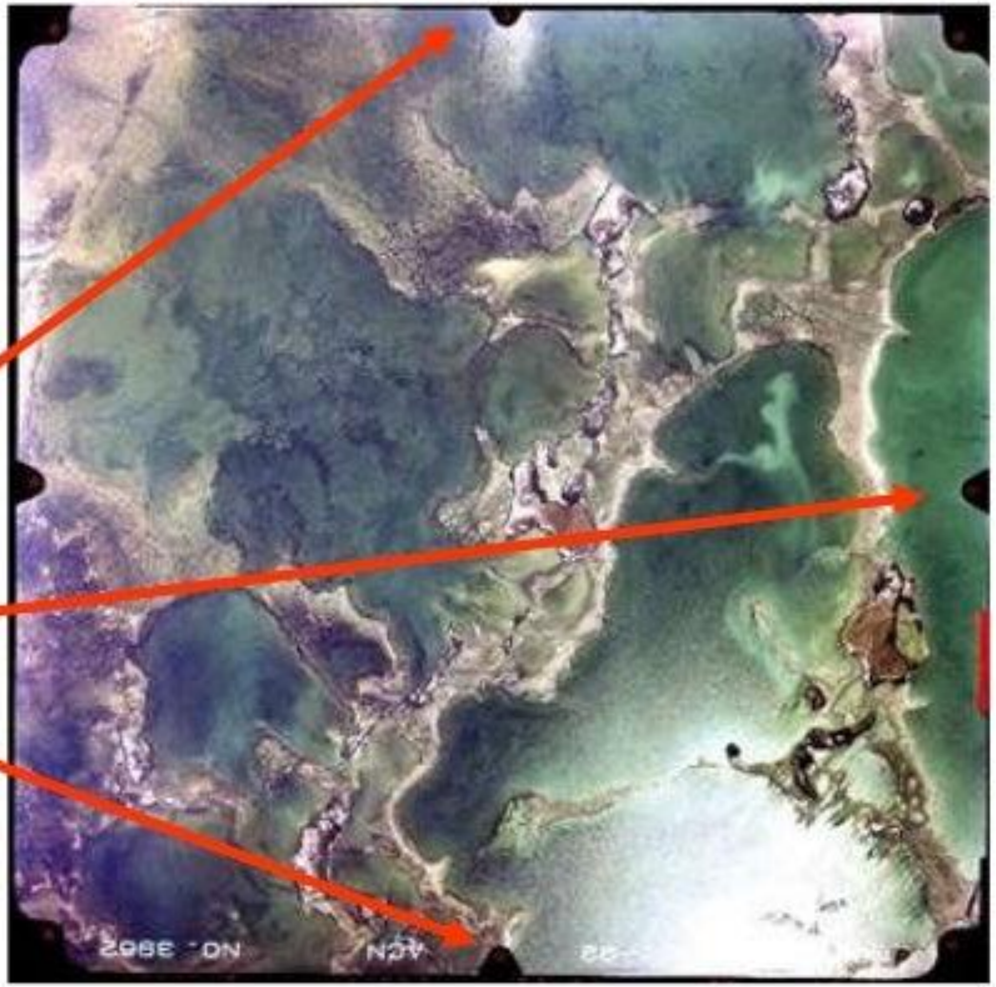
When you know the scale you can take 2-D measurements from a photograph
(e.g.
horizontal distance, horizontal area, etc.)

Terminology

Fiducial marks is a set of marks located in the corners or edge-centers, or both, of an aerial photographic image.

These marks are exposed within the camera onto the original film and are used to define the frame of reference for spatial measurements on aerial photographs.

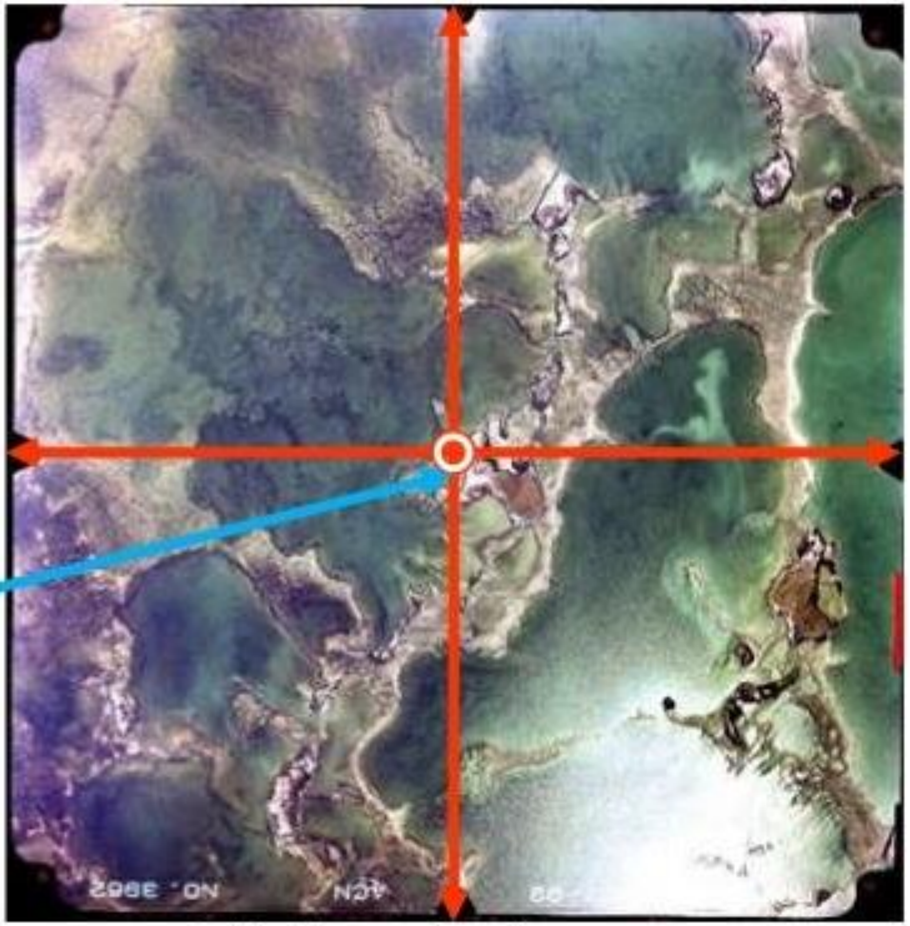
Fiducial marks



Terminology

Opposite fiducial marks connected, intersect at approximately the image center or principal point of the aerial photograph.

The principal point is the geometric center of the photograph



Principal point

Fiducial

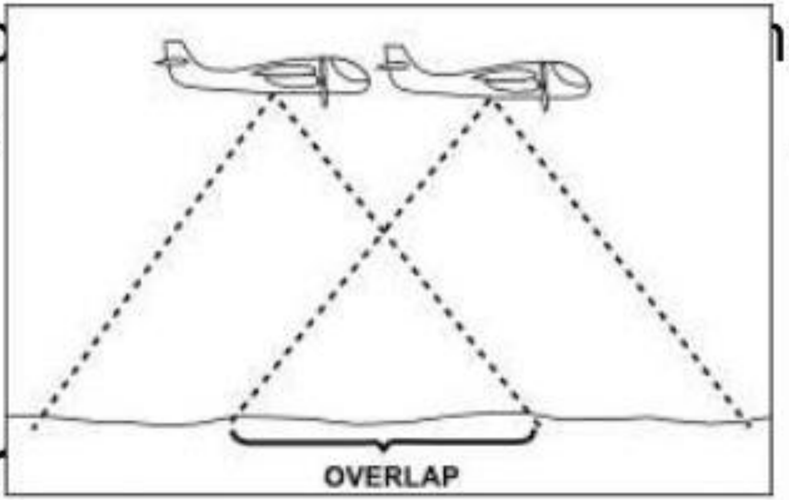
Flight Planning

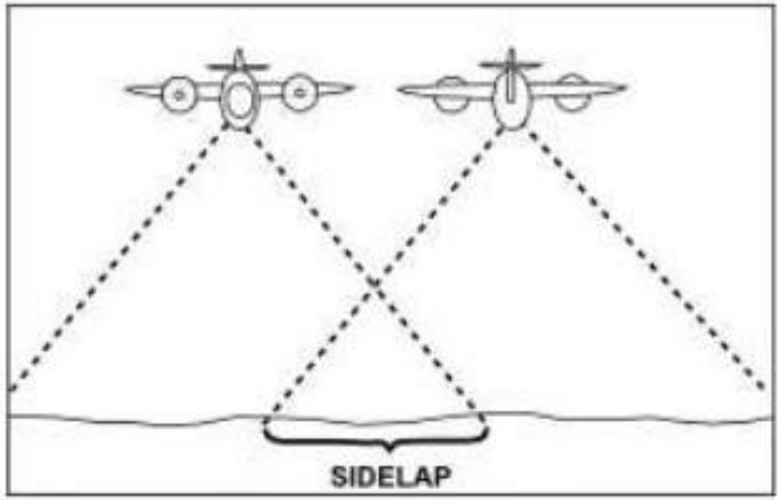
Aerial photo projects for all mapping and most image analyses require that a series of exposures be made along each of the multiple flight lines.

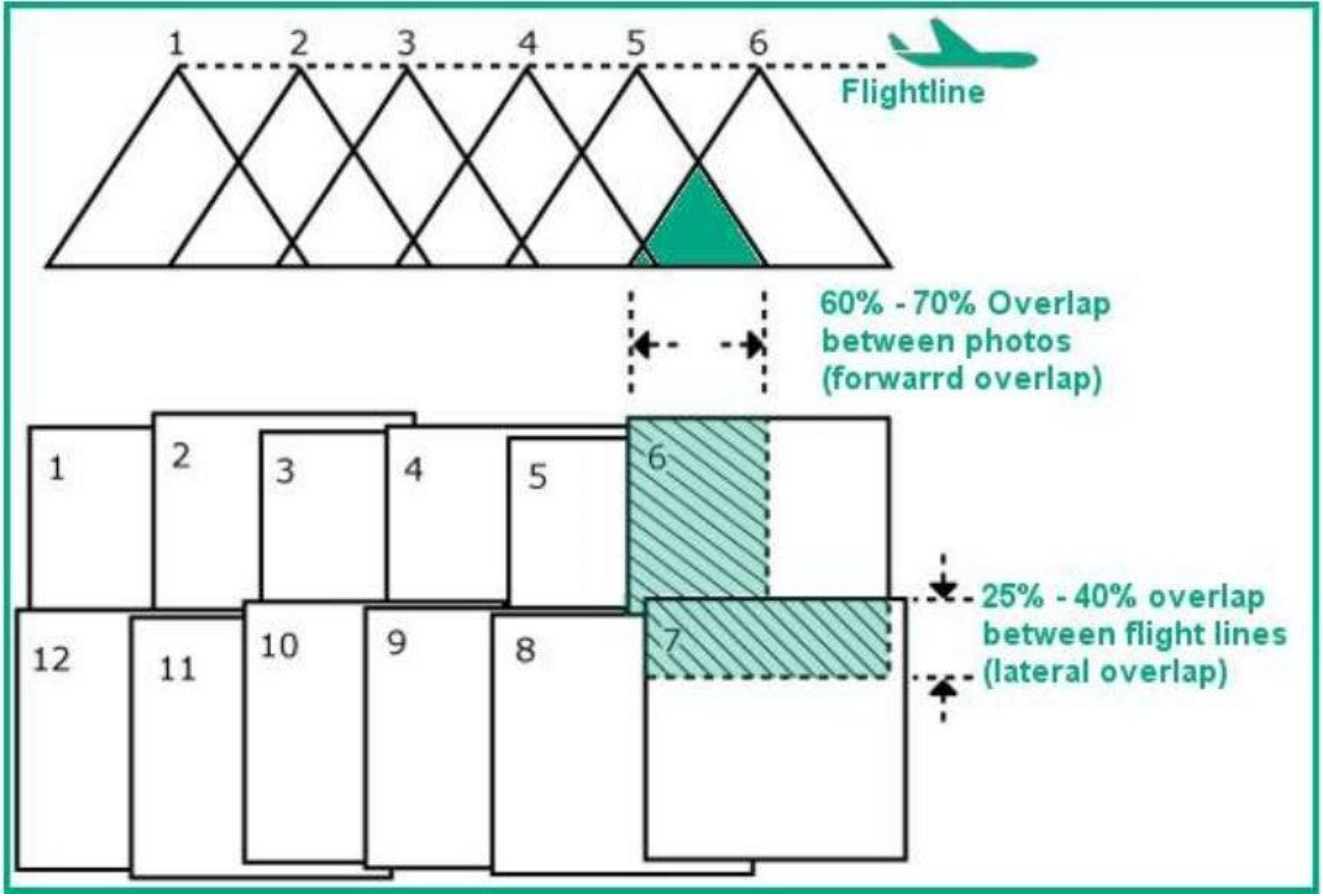
To guarantee stereoscop hout the site, the photographs must overlap

a) **in the line of flight**
(Overlap)
needed for parallax

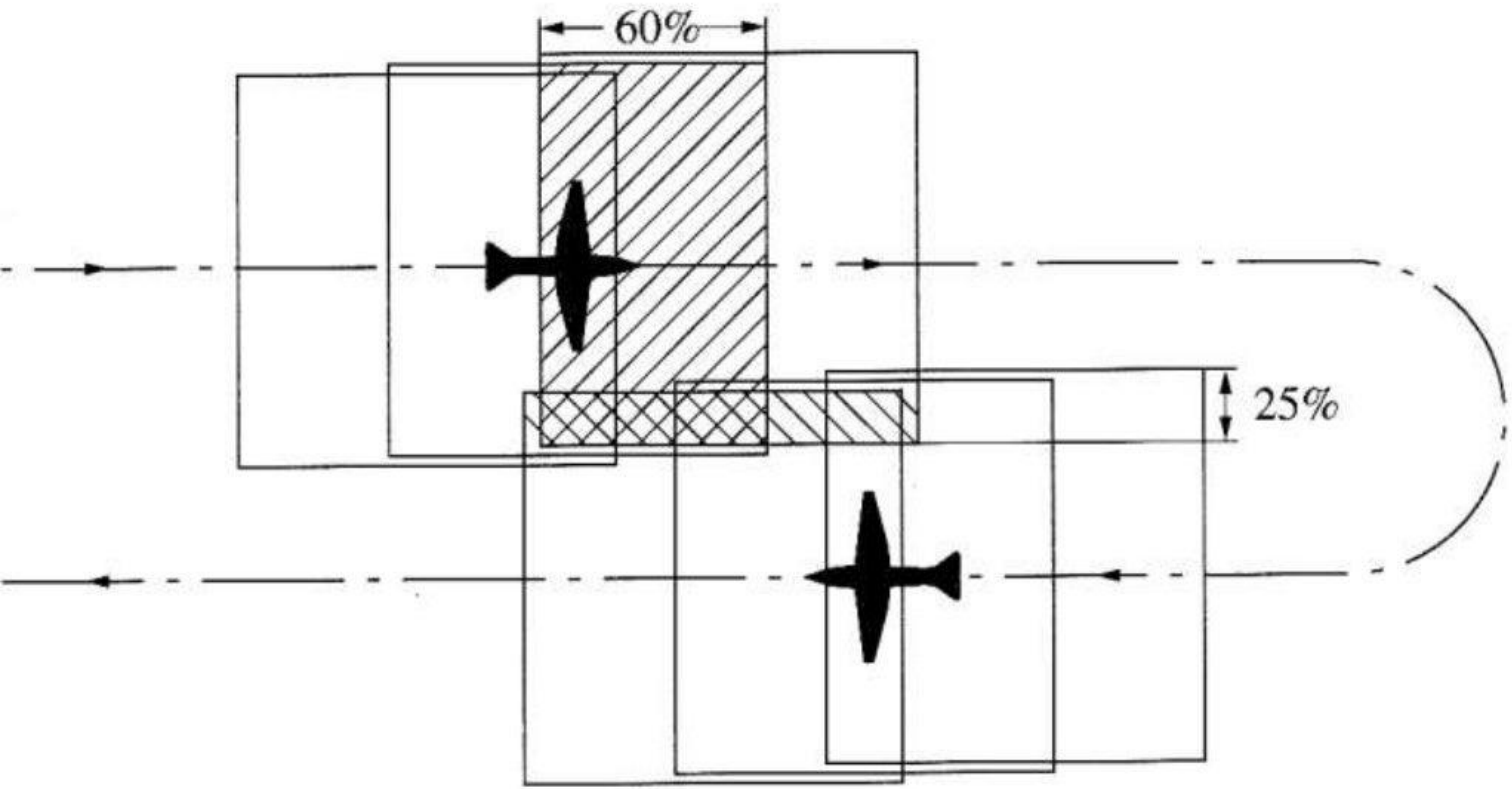
a) **between adjacent fligh**
(Sidelap)
to avoid missing bits

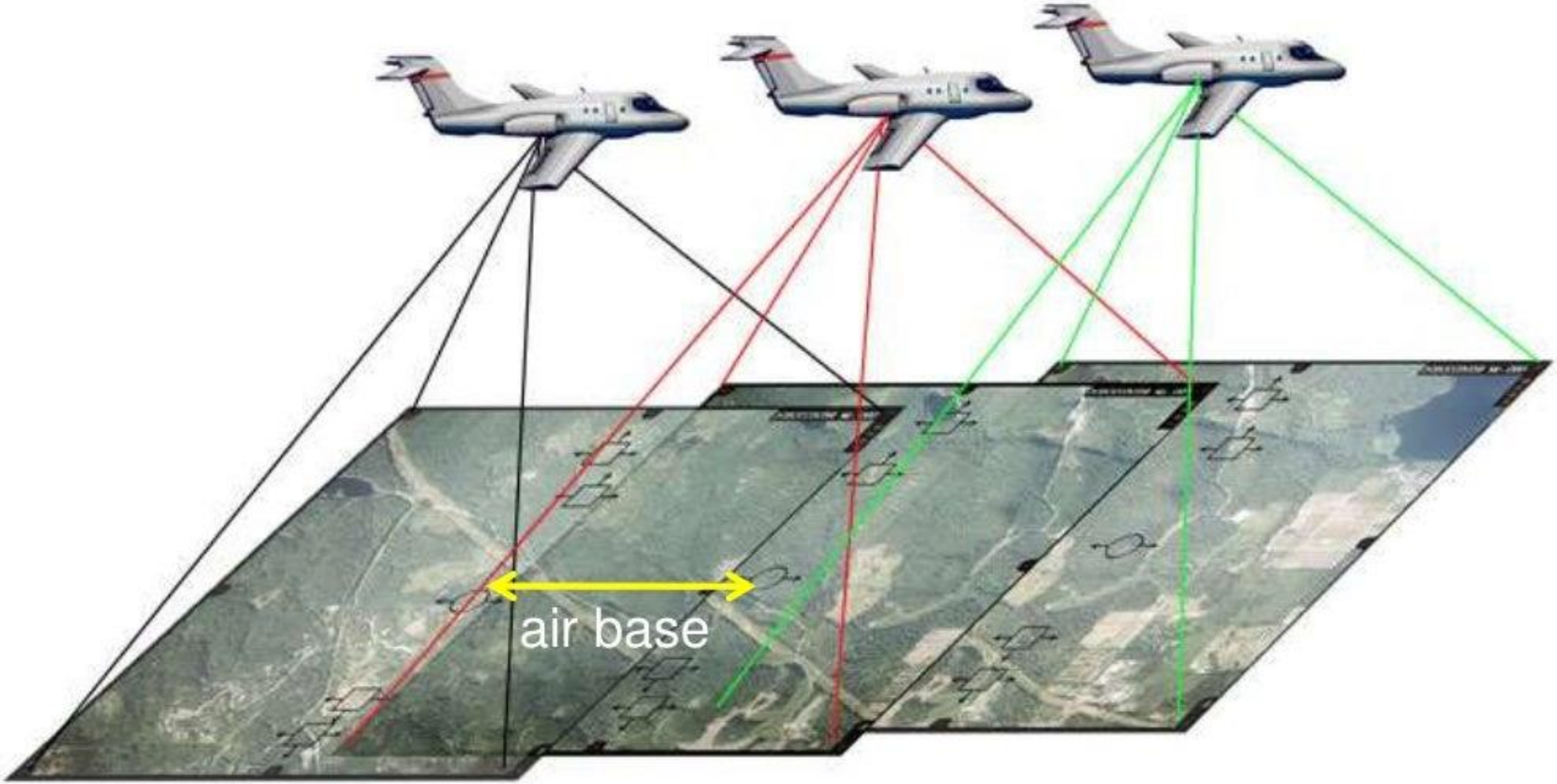






1
2
3
4
5
6
Flightline
60% - 70% Overlap
between photos
(forwarrd overlap)
1
2
3
4
5
6
12
11
10
9
8
7
25% - 40% overlap
between flight lines
(lateral overlap)


60%
25%



Distance between principal point of adjacent photographs is known as the **"air base"**

Fundamentals of Stereoscopy

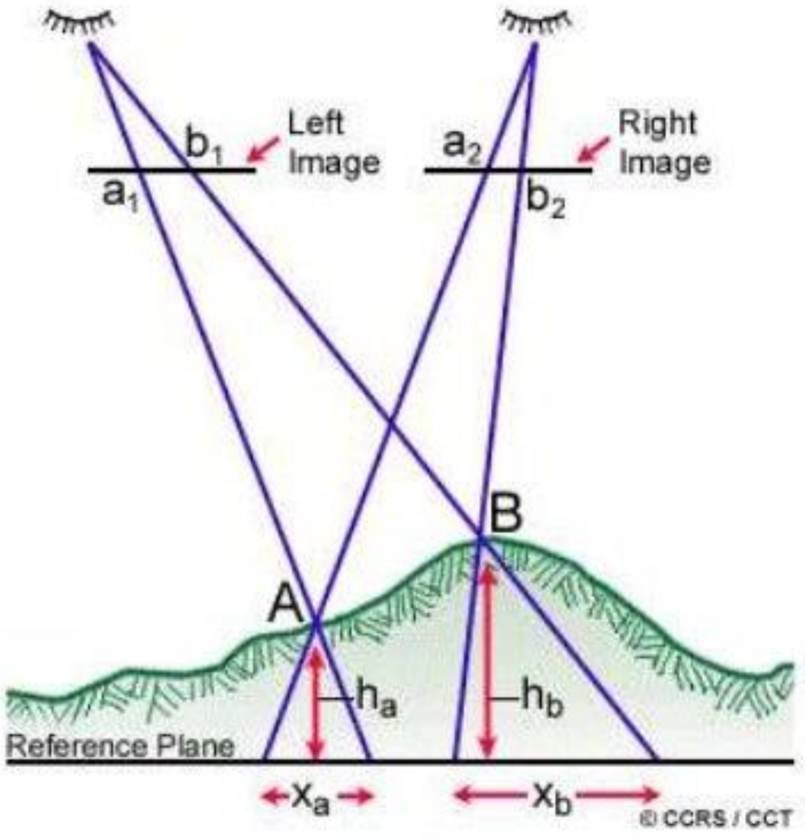
Stereoscopy, sometimes called stereoscopic imaging, is a technique used to enable a three-dimensional effect, adding an illusion of depth to a flat image.

In aerial photography, when two photographs overlap or the same ground area is photographed from two separate position forms a **stereo-pair**, used for three dimension viewing. Thus obtained a pair of
stereoscopic photographs or images can be viewed stereoscopically to determine parallax and stereo/3D viewing

Parallax Measurement

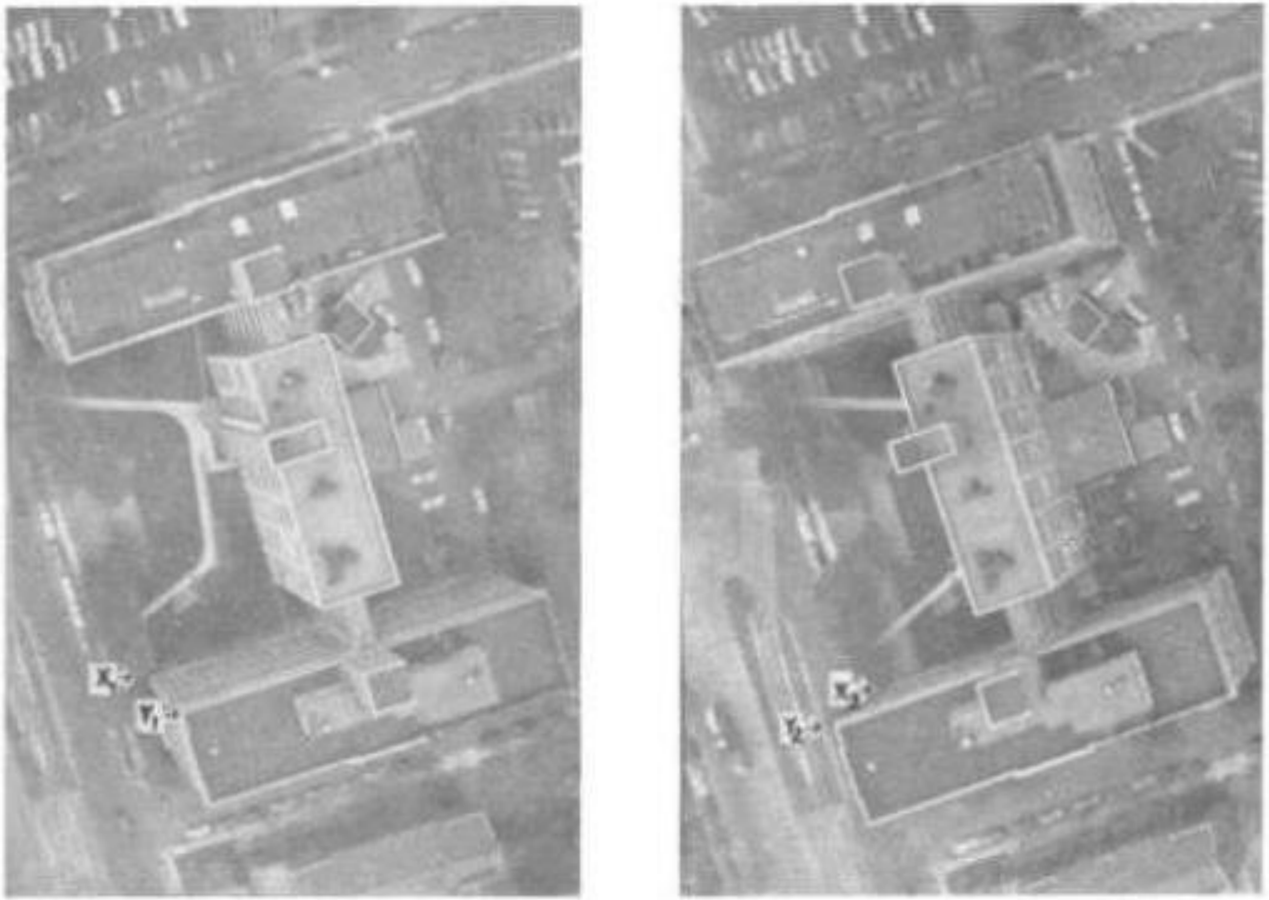
Parallax:
The displacement of an object caused by a change in the point of observation is called parallax.

Stereoscopic parallax is caused by taking photographs of the same object but from different points of observation.



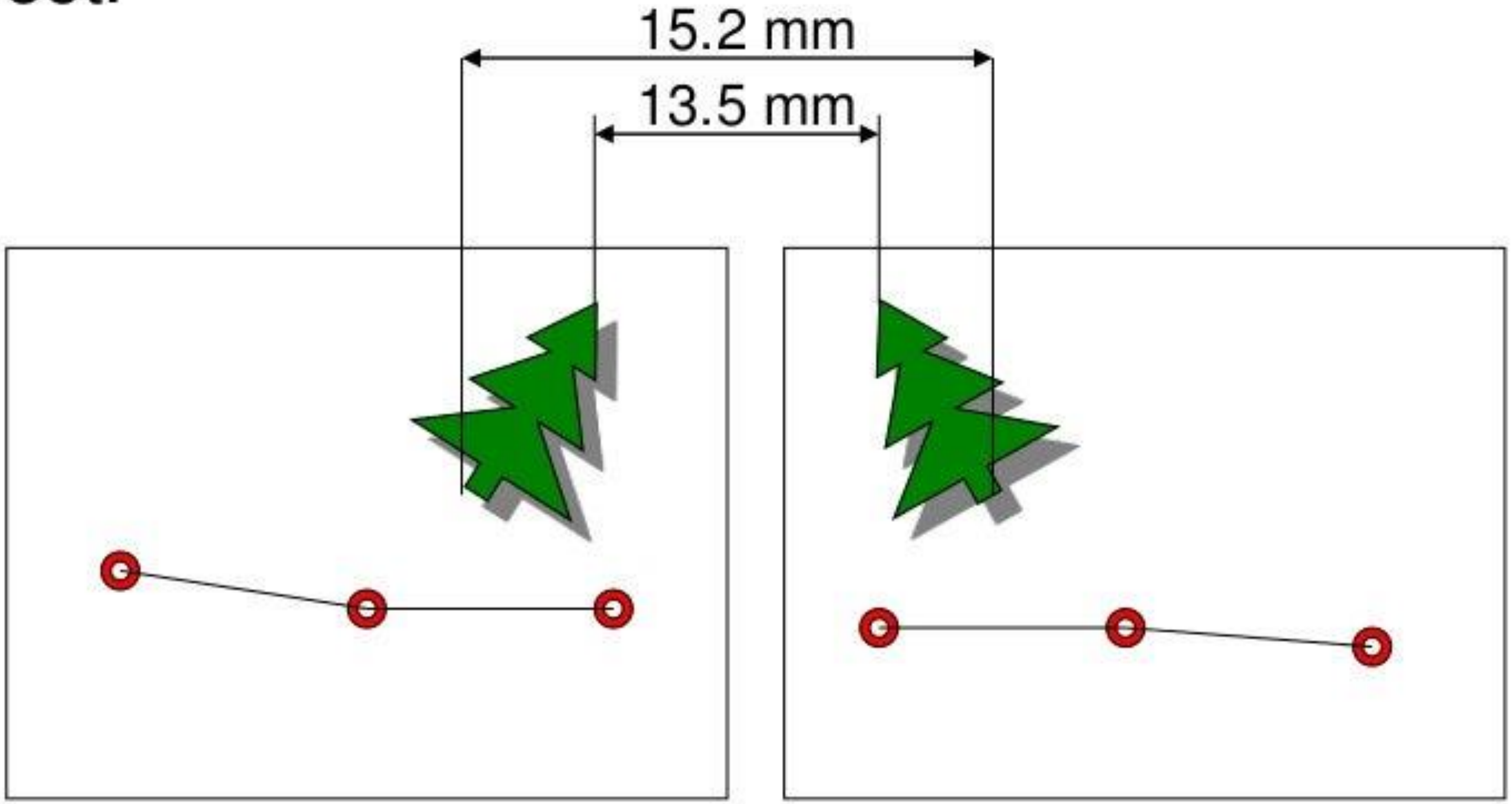
A stereoscope is a device for **viewing a stereoscopic pair of separate images**, depicting left-eye and right-eye views of the same scene, as a single three-dimensional image.

Stereoscopic parallax



The same principle can be used to find height of objects in stereopairs of vertical aerial photographs

- **Differential Parallax** is the difference between the stereoscopic parallax at the top and base of the object.



dP = 15.2mm – 13.5mm = 1.7 mm

Parallax Measurement

$$\text{Height of object} = \frac{H \cdot dP}{P + dP}$$

H = height of aircraft above ground
P = absolute parallax at base of object being measured*
dP = differential parallax

* For convenience the photo base length of a stereo pair is commonly substituted for absolute stereoscopic

Difference Between Map & Aerial Photograph

Point	Map	Aerial Photograph
Preparation	Maps are prepared by actually taking the linear and angular measurements of ground or terrain	A photograph of the terrain or ground are taken by a areal camera show all the details is know as aerial photograph
Time	Need more time to prepare a map	Need less time to prepare a photograph
Methods	Maps can be prepare by conventional methods of surveying in which less skilled person is required	Aerial photographs are prepared by aerial surveying which is high skilled jobs and also need highly skilled technicians
Confusing details	Maps are prepared by conventional method which do not depicts confusing details and easily recognizable	Aerial photographs depict confusing details which are not easily identified.
Distortion	There is no distortion caused by tilt or relief displacement because maps are prepared after necessary measurements	There is distortion of details of terrain or ground caused by tilt and relief displacement
Cost	Plotting of map is economic	Preparation of maps from aerial photograph is not economical
Reading	Reading of map is common and more familiar to users. Different detail of different objects like roads pond rivers, railway, crossing, etc can be shown by conventional sign and symbols	The picture of aerial photographs are not familiar with the eyes of user and it need a special techniques to read the photo views
Scale	The scale of maps is uniform through the map extent	The scale of the photograph is not uniform
Projection	It is an orthogonal projection	It is a central projection
Area coverage	Mapping of inaccessible area is very difficulty and sometimes it becomes impossible	Aerial photography of inaccessible area is possible.

References :

1) Text book of Remote Sensing and Geographical Information System by M.Anji Reddy by BS Publications.
2) Introduction to photogrammetry by Dr. Rambabu Palaka, Associate Prof. BVRIT